

MAYILRAJ R

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SUMMARY

Final-year B. Tech (IT) student with full-stack development experience across two internships, building tested REST APIs, React dashboards, and Django/Flask backends. Strong focus on OOP/OOAD design, unit and integration testing, and measurable performance gains (reduced latency, cut rollbacks, improved test coverage).

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), Embedded C, SQL · Data Structures & Algorithms

Backend / APIs: Node.js, Express.js, Flask, Django · REST API Design, Unit & Integration Testing (pytest, Jest)

Frontend: React.js, HTML5, CSS3, Bootstrap 5 · Chart.js data visualisation

Databases: MySQL, PostgreSQL, MongoDB/Mongoose

Tools & Practices: Git/GitHub, Postman, VS Code · Agile · OOP/OOAD · CI/CD pipelines

EXPERIENCE

Software Development Engineer Intern | Ibee Analytics, Hyderabad Aug 2025 – Sep 2025

- Designed clean, modular Python backend modules using OOP/OOAD principles and wrote pytest unit tests achieving >90% regression coverage across 5+ API endpoints, reducing post-release bug discovery by ~30%.
- Built and shipped 3 scalable RESTful APIs with Node.js/Express.js following modular MVC architecture; performed impact analysis before each sprint release, cutting unplanned rollbacks to zero.
- Developed React.js analytics dashboards integrated with Chart.js, visualising real-time KPIs for business stakeholders; optimised MongoDB schemas (compound indexes, projection), reducing average read latency by ~25%.
- Collaborated across 4 Agile sprints using Git branching and PR-based code reviews, maintaining 100% on-time milestone delivery.

Web Developer Intern | Hapus Infotech, Coimbatore Jul 2025 – Aug 2025

- Refactored legacy Python/Django codebase by applying OOP design patterns (Factory, Strategy), cutting average page-load time by ~20% and reducing duplicated code by 35%.
- Implemented JWT-based user authentication and session management with input validation; designed responsive UI using Bootstrap 5, raising mobile usability score from 62 to 87 (Lighthouse).

PROJECTS

Pranara Tourism – Full-Stack Travel Platform & Technical SEO | React 19, Vite, Python (Flask), Firebase, Supabase

- Engineered and deployed a production travel platform (pranaramunnar.com) using React 19, Vite, Flask REST APIs with Firebase Authentication and Supabase integration.
- Executed a technical SEO overhaul, resolving 7+ audit failures and raising the site's SEO score from 74/100 to near-perfect indexability by fixing H1/H2 structure, canonical tags, OpenGraph/Twitter cards, and JSON-LD Schema.org data.
- Optimised Core Web Vitals by eliminating render-blocking scripts and building pre-rendered semantic HTML fallbacks for search-engine crawlers, achieving sub-0.8s TTFB and improved LCP/CLS.

Empower Her – Safe Route Navigation | Android Studio, Google Maps SDK, Firebase, AI/ML

- Designed OOP architecture for an Android safety app with an AI-based threat detection module; wrote unit tests for GPS parsing and real-time alert logic, achieving zero false-negative alerts in testing.
- Integrated Firebase Realtime Database for <200ms GPS-sharing latency; implemented smart-ring SOS trigger via BLE, enabling one-tap emergency broadcast to 5 pre-configured contacts.

Predictive Marketing Dashboard | Python, Flask, Facebook Prophet, Beautiful Soup, REST API

- Architected a modular, scalable Flask backend using OOAD patterns (Repository, Service Layer), enabling independent unit testing of each component with pytest; achieved 95%+ test coverage on the forecasting pipeline.
- Integrated Facebook Prophet time-series model to forecast marketing KPIs with <8% MAPE on validation data; automated SEO metric collection via Beautiful Soup scrapers exposed as REST endpoints.
- Delivered iterative milestones with documented impact analysis; the dashboard reduced manual reporting effort by an estimated 4 hours/week for the client team.

EDUCATION

B. Tech – Information Technology, University College of Engineering, Tiruchirappalli

Aug 2023 – May 2026 · CGPA: 7.6 / 10

Diploma – Electronics & Communication Engineering, Thevanesam Eruthaya Ammal Polytechnic College

Jun 2020 – Apr 2022 · CGPA: 9.1 / 10

CERTIFICATIONS & ACHIEVEMENTS

- Deloitte Cyber Job Simulation (Forage) — threat analysis, SIEM triage, and incident reporting workflows.
- TATA Cybersecurity Analyst Simulation (Forage) — IAM, network defence, and vulnerability assessment modules.
- Python Programming – GUVI (HCL) · SQL Bootcamp – Lets Upgrade · Java Training – LinkedIn Learning
- MeitY – ISEA Digital Security Quiz, Ministry of Electronics & IT, Government of India